

Geometrical Optics, solved

Step-by-step solutions to every practice problem in Chapter 22. Try each problem on your own before you read its solution.

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P22.1

Rays from the submerged part of the straw travel up through the water and refract **away from the normal** as they cross into the less dense air. Your eye traces the emerging rays straight back, so they seem to come from a point *higher* than the straw really is: the apparent depth is about d/n . The submerged part therefore appears raised towards the surface, making the straw look bent, or broken, at the waterline.

Ray diagram: draw a ray from the bottom of the straw to the surface. At the surface it bends away from the normal towards your eye. Extend it backwards in a straight line, and it meets the vertical above the true end of the straw. That point is where the end appears to be.

P22.2

Diamond's refractive index (2.42) is much higher than glass's (about 1.5), so its critical angle is only 24.4° , compared with 41.8° for glass. A brilliant cut is designed so that light entering through the top strikes the back facets beyond this small critical angle and is totally reflected, back up through the top towards the viewer. Glass, with its larger critical angle, lets much of the light leak out of the back. Diamond also has strong **dispersion**, so it splits white light into flashes of colour ("fire").

P22.3

No. For a convex mirror $f < 0$. With a real object, $s > 0$, so

$$\frac{1}{s'} = \frac{1}{f} - \frac{1}{s} < 0 \Rightarrow s' < 0.$$

The image is always virtual (behind the mirror), upright and smaller. A convex mirror spreads light out, so reflected rays never actually converge.

P22.4

About two-thirds of the eye's focusing power comes from refraction where light passes from air ($n = 1.00$) into the cornea ($n = 1.376$). Underwater, the outside medium has $n = 1.333$, so the index jump at the cornea almost disappears, and with it most of the refraction (Problem P22.14). The eye becomes very long-sighted, and images focus far behind the retina, so everything blurs.

Goggles hold a layer of air against the cornea, which restores the air–cornea interface. Their flat windows add essentially no power, so the eye focuses normally.

P22.5

Refraction into the slab:

$$\sin \theta_2 = \frac{\sin 60^\circ}{1.50} = 0.577 \Rightarrow \theta_2 = 35.3^\circ.$$

Emerging ray: at the parallel back face, Snell's law in reverse gives 60° again. The ray emerges parallel to the incident ray, but shifted sideways.

Sideways displacement. The path inside the slab has length $t / \cos \theta_2$, and the emerging ray is offset from the original line by this length times $\sin(\theta_1 - \theta_2)$:

$$d = \frac{t \sin(\theta_1 - \theta_2)}{\cos \theta_2} = \frac{5.0 \sin 24.7^\circ}{\cos 35.3^\circ} = \frac{5.0(0.419)}{0.816} = 2.6 \text{ cm.}$$

P22.6

Without cladding, with air outside:

$$\text{NA} = \sqrt{1.62^2 - 1.00^2} = \sqrt{1.62} = 1.27.$$

A value above 1 simply means that *every* ray entering the end face is trapped.

Why cladding is still essential. Total internal reflection needs a clean boundary with a lower-index medium. On a bare fibre, scratches, dust, moisture, fingerprints or contact with the holder or a neighbouring fibre in a bundle would locally destroy that boundary. Light would then leak

out (and, in an imaging bundle, leak into the neighbouring fibres and blur the image). The cladding provides a permanent, protected interface, and it keeps the guided light away from the outer surface.

P22.7

An upright, magnified image needs a **concave** mirror, with the face inside the focal length. With $m = +2$ and $s = 15$ cm, $s' = -30$ cm (a virtual image behind the mirror).

$$\frac{1}{f} = \frac{1}{15} + \frac{1}{-30} = \frac{1}{30} \Rightarrow f = 30 \text{ cm}, \quad R = 2f = 60 \text{ cm}.$$

P22.8

$f = R/2 = -0.50$ m (convex) and $s = 10$ m:

$$\frac{1}{s'} = -\frac{1}{0.50} - \frac{1}{10} = -2.1 \text{ m}^{-1} \Rightarrow s' = -0.48 \text{ m}, \quad m = -\frac{-0.476}{10} = +0.048.$$

The image is virtual, upright, and about 1/21 of full size, just behind the mirror.

P22.9

(a) $s = 15$ cm:

$$\frac{1}{s'} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30} \Rightarrow s' = 30 \text{ cm}, \quad m = -2.0.$$

Real, inverted, twice the size, 30 cm beyond the lens. In the ray diagram, the object lies between F and 2F, and the rays converge beyond 2F'.

(b) $s = 5.0$ cm:

$$\frac{1}{s'} = \frac{1}{10} - \frac{1}{5.0} = -\frac{1}{10} \Rightarrow s' = -10 \text{ cm}, \quad m = +2.0.$$

Virtual, upright, twice the size, on the same side as the object. In the ray diagram, the rays diverge after the lens, and their backward extensions meet. This is how a magnifying glass works.

P22.10

$$P = \frac{1}{0.20} + \frac{1}{-0.30} = 5.0 - 3.33 = +1.7 \text{ D}, \quad f = \frac{1}{1.67} = 0.60 \text{ m} = 60 \text{ cm}.$$

The combination is **converging** (positive power), because the converging lens is the stronger of the two.

P22.11

The lens must take an object at $s = 0.25 \text{ m}$ and form a virtual image at the near point, $s' = -0.75 \text{ m}$:

$$P = \frac{1}{0.25} + \frac{1}{-0.75} = 4.0 - 1.33 = +2.7 \text{ D}.$$

Typical “reading glasses” for this age are about +2.5 D.

P22.12

(a) Image at infinity: $M = N/f = 25/5.0 = 5.0$.

(b) Image at the near point: $M = 1 + N/f = 6.0$.

P22.13

Set up. Light travels from the fish, through the water ($n_1 = 1.333$), out into the air ($n_2 = 1.00$). The fish is $s = 10 \text{ cm}$ from the front surface. The surface is convex towards the observer, and its centre of curvature lies inside the bowl, on the *incoming* side. In the real-is-positive convention, that makes $R = -15 \text{ cm}$.

Execute:

$$\frac{n_1}{s} + \frac{n_2}{s'} = \frac{n_2 - n_1}{R} \Rightarrow \frac{1.333}{10} + \frac{1}{s'} = \frac{-0.333}{-15}.$$

$$\frac{1}{s'} = 0.0222 - 0.1333 = -0.111 \text{ cm}^{-1} \Rightarrow s' = -9.0 \text{ cm}.$$

The image is virtual, 9.0 cm behind the front surface, so the fish appears about 1 cm closer than it really is.

Magnification:

$$m = -\frac{n_1 s'}{n_2 s} = -\frac{1.333(-9.0)}{1.00(10)} = +1.2.$$

The fish looks upright and 20% larger than it is. The curved bowl acts as a weak magnifying lens.

P22.14

(a) **In air:**

$$P = \frac{n_2 - n_1}{R} = \frac{1.376 - 1.000}{0.0078} = 48 \text{ D.}$$

(The back surface of the cornea has a small negative power, which brings the cornea's total to about 43 D.)

(b) **Underwater:**

$$P = \frac{1.376 - 1.333}{0.0078} = 5.5 \text{ D.}$$

(c) **Lost:** $48.2 - 5.5 \approx 43$ D, about 70% of the eye's total power of about 60 D. Images now focus far behind the retina, so the eye becomes extremely long-sighted, and the lens's accommodation (at most about 10 D in a young person) cannot make up the difference. Goggles put air back in front of the cornea and restore the full 48 D. Diving animals have flattened corneas and very powerful, spherical lenses for exactly this reason.

P22.15

(a) **Intermediate image.** For the final image to be at infinity, the intermediate image must lie at the eyepiece's focal point, 2.5 cm in front of the eyepiece. That is $22.1 - 2.5 = 19.6$ cm behind the objective, so $s'_o = 19.6$ cm.

(b) **Object position:**

$$\frac{1}{s_o} = \frac{1}{1.6} - \frac{1}{19.6} = 0.625 - 0.051 = 0.574 \Rightarrow s_o = 1.74 \text{ cm,}$$

just outside the objective's focal point.

(c) Magnification:

$$m_o = -\frac{19.6}{1.74} = -11.3, \quad M_e = \frac{25}{2.5} = 10, \quad M = (-11.3)(10) \approx -113.$$

The approximate formula gives $-(L/f_o)(N/f_e) = -(18/1.6)(10) = -112$, where $L = 22.1 - 1.6 - 2.5 = 18$ cm is the distance between the focal points. It agrees well.

P22.16

(a) Refractive index:

$$n = \frac{\sin \frac{1}{2}(60^\circ + 37.2^\circ)}{\sin 30^\circ} = \frac{\sin 48.6^\circ}{0.500} = \frac{0.750}{0.500} = 1.50.$$

(b) Red and violet. Rearranging, $\sin \frac{1}{2}(A + \delta) = n \sin 30^\circ = n/2$:

- Red: $\sin \frac{1}{2}(60^\circ + \delta) = 0.7565$, so $\frac{1}{2}(60^\circ + \delta) = 49.16^\circ$ and $\delta_{\text{red}} = 38.3^\circ$.
- Violet: $\sin \frac{1}{2}(60^\circ + \delta) = 0.766$, so $\frac{1}{2}(60^\circ + \delta) = 50.0^\circ$ and $\delta_{\text{violet}} = 40.0^\circ$.

Angular spread: about 1.7° . This is enough to spread white light into a visible spectrum on a screen a metre or so away, which is the principle of the prism spectroscope.

P22.17

Lens 1:

$$\frac{1}{s'_1} = \frac{1}{10} - \frac{1}{30} = \frac{1}{15} \Rightarrow s'_1 = 15 \text{ cm}, \quad m_1 = -\frac{15}{30} = -0.50.$$

Lens 2. The first image is $25 - 15 = 10$ cm in front of lens 2, and it serves as a real object for it, so $s_2 = 10$ cm:

$$\frac{1}{s'_2} = \frac{1}{15} - \frac{1}{10} = -\frac{1}{30} \Rightarrow s'_2 = -30 \text{ cm}, \quad m_2 = -\frac{-30}{10} = +3.0.$$

Final image: virtual, 30 cm in front of lens 2, which is 5 cm in front of lens 1, on the object's side. The overall magnification is

$$m = m_1 m_2 = (-0.50)(3.0) = -1.5,$$

so the image is inverted relative to the object and 1.5 times larger. Lens 2 is acting as a magnifier for the real image formed by lens 1, which is the principle of the compound microscope.

P22.18

(a) **Power of the corneal surface:**

$$P = \frac{1.376 - 1}{0.00780} = 48.2 \text{ D.}$$

(b) **New radius.** The surface must lose 2.5 D, so its power must become 45.7 D:

$$R' = \frac{0.376}{45.7} = 8.23 \text{ mm.}$$

The centre of the cornea must be *flattened*.

(c) **Change in radius:** $8.23 - 7.80 = 0.43$ mm, which is about 430 μm .

What the laser actually removes. Over the central optical zone (about 6 mm across, so 3 mm in radius), the change in the surface's sag (its depth below a flat plane) is

$$\frac{a^2}{2R} - \frac{a^2}{2R'} = \frac{9}{2(7.80)} - \frac{9}{2(8.23)} = 0.577 - 0.547 = 0.030 \text{ mm} = 30 \mu\text{m.}$$

That is only about 12 μm per dioptre. Each excimer laser pulse removes roughly 0.25 μm of tissue. An error of a few micrometres in depth would leave a noticeable residual error, which is why LASIK systems use eye-tracking and very precisely calibrated, computer-controlled laser pulses.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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